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Techno-Economic and Environmental Analysis and Evaluation of 40 MW Solar Photovoltaic Projects in Iraq

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Abstract: This study investigates the technical and financial viability of a proposed 40 MW solar photovoltaic (PV) power plant in two Iraqi regions: Al Sulaymaniyah and Al Muthana. The study encompassed technical aspects, economic assessment, cost analysis, and estimation of net annual greenhouse gas (GHG) reduction. Emphasis was placed on optimization of energy production and cost-effectiveness. The analysis utilizes the RETScreen software to assess the feasibility of installing PV systems at the selected sites. The research parameters considered in the study included the horizontal and tilted daily solar radiation, the annual electricity production, capacity factor, and GHG reduction. The financial analysis and feasibility included simple payback (SPB), net present value (NPV), and annual life cycle savings. These projects can be best achieved when supported by grants and a reasonable tariff starting from \$0.035/kWh or more, resulting in a NPV of \$5,209,217 and \$4,819,530 in the Al Sulaymaniyah and Al Muthana projects, respectively.

Keywords: Electric energy production; Power plant; Solar photovoltaic system; RETScreen

1 Introduction

The significant need to treat climate change has hastened the global shift toward sustainable, low-carbon energy alternatives. Among these, solar photovoltaic (PV) technology stands out as a top solution, thanks to its scalability, rapidly decreasing costs, and low environmental impact during operation [1]. Al-Kayiem and Mohammad [2] reported an outlook on the energy scenario in Iraq, emphasizing the potential of solar power production. They concluded that solar-based power production is highly suitable for Iraq due to its favorable climate. However, they included the challenges facing the country's adoption of solar-based power production. As nations strive to meet commitments under the agreements and achieve net-zero emissions targets, large-scale solar PV projects, such as PV plants with different power capacities, offer a practical and impactful pathway to decarbonize electricity generation [3]. Tezer et al. [4] presented a comprehensive evaluation of the various optimization approaches applied to stand-alone hybrid renewable energy systems (HRES). The authors review a wide range of optimization methods, including classical, heuristic, and hybrid techniques, highlighting their strengths, limitations, and suitability for different system configurations and objectives. By systematically comparing these approaches, the paper provides valuable insights for researchers and practitioners in selecting appropriate optimization strategies tailored to specific HRES projects. The findings underscore the importance of multi-objective optimization and suggest future directions for enhancing optimization frameworks to better address the complexity and dynamics of renewable-based energy systems. Currently, the majority of newly installed global electricity capacity comes from renewable sources, with solar and wind power experiencing substantial cost reductions and becoming increasingly competitive with traditional fossil fuels [5, 6].

Solar PV plants represent a mid-to-large-scale utility installation capable of producing substantial clean energy, often enough to power tens of thousands of households annually. In addition to supplying renewable electricity, such projects can play a pivotal role in displacing fossil fuel-based power generation, thereby significantly reducing greenhouse gas (GHG) emissions. However, the successful deployment of these systems depends on various factors, including technical viability, economic competitiveness, policy support, environmental impact, and grid integration capabilities [7, 8]. Khalid [9] analyzed the performance of large-scale solar PV installations in high-irradiance, arid environments. Using PVsyst simulations, the results show that plants can achieve a capacity factor of 20–23%, with optimal output influenced by tilt angle and temperature derating. Technical feasibility is strong, provided dust management systems are in place.

Most researchers have evaluated the feasibility and effectiveness of solar PV projects in reducing GHG emissions. Leewiraphan et al. [10] demonstrate that under stable financial and regulatory conditions, the levelized cost of electricity (LCOE) for 40 MW projects can be as low as $0.05/kWh, thereby confirming economic competitiveness. Nugent and Sovacool [11] provide a lifecycle emissions analysis, indicating emissions as low as 18–25 $\mathrm{g\,CO_2e/kWh}$, which is significantly lower than those from fossil-based sources. Integration challenges are explored by Shafiullah et al. [12], who emphasize the role of smart inverters and energy storage in grid stability. Tsoutsos et al. [13] assess environmental trade-offs, recommending low-impact land-use practices, such as agrivoltaics, to reduce ecological disruption. Van Opstal and Smeets [14] emphasize the importance of robust policy frameworks, such as feed-in tariffs and renewable auctions, in promoting PV adoption. Virah-Sawmy and Sturmberg [15] analyze the socioeconomic effects of solar PV in emerging economies, noting positive outcomes like local job creation and improved energy access. Akhtar et al. [16] compare the feasibility of emissions reduction in India and South Africa, confirming the regional potential for projects to exceed 35,000 tons of $\mathrm{CO_2}$ annually. Deshmukh et al. [17] examine the operational performance of existing PV systems, reporting long-term reliability and low degradation rates. Ajel et al. [18] modelling results show how scaling PV can significantly contribute to climate goals, potentially abating over 2 million metric tons of $\mathrm{CO_2}$ per year. Deshmukh et al. [17] and Ajel et al. [18] emphasized the role of simulation tools, such as Hybrid Optimization Model for Multiple Energy Resources (HOMER) and RETScreen, in the design and evaluation of hybrid energy systems to enable stakeholders to make suitable decisions regarding the hybrid energy projects. Zubi et al. [19] conducted a RETScreen-based assessment of solar energy projects in the United Arab Emirates (UAE), focusing on technical performance, economic viability, and environmental impact. Their findings demonstrated that solar PV systems in the UAE are not only technically feasible and economically attractive but also offer considerable potential for reducing carbon emissions. The study provided valuable insights to policymakers and investors for expanding the country's renewable energy portfolio.

The existing literature consistently affirms that solar PV projects are technically sound, financially viable, and environmentally essential for achieving sustainable energy transitions. While previous studies have investigated various implementation strategies, the detailed technical and financial feasibility of establishing a utility-scale solar PV plant remains underexplored, particularly in the context of Iraq. This study addresses that gap by evaluating the feasibility of a 40 MW solar PV power plant in two Iraqi locations, Al Sulaymaniyah and Al Muthana. Using RETScreen-based analysis, the research focuses on optimizing both energy output and cost efficiency, providing a comprehensive assessment to support future investment and policy development in Iraq's renewable energy sector.

With its abundant solar radiation and extensive area availability, Iraq is well-positioned to utilize solar energy as a vital component of its energy mix. Investing in solar plants can address the country's rising electricity demand, reduce dependence on oil-based fuels, lower GHG emissions, and stimulate economic growth through job creation and technological advancements. Embracing solar power presents Iraq with a pathway toward a cleaner, more resilient energy future that aligns with its long-term environmental and economic objectives. Achieving this vision will depend on robust government support, well-crafted policy frameworks, and active collaboration with international partners to overcome challenges and ensure long-term sustainability.

2 Materials and Methods

Iraq possesses significant potential for solar energy due to its geographical location, which is characterized by high solar irradiance and long hours of sunlight throughout the year. Hence, the adoption of PV power plants in Iraq is driven by several key motivations.

• Abundant solar resources: Iraq receives an average of 5–7 $\mathrm{kWh/m^2}$ of solar energy daily, making it one of the most solar-rich countries in the region. This natural advantage provides an opportunity to harness clean and sustainable energy efficiently.

• Energy Diversification: Iraq's current energy mix relies heavily on finite fossil fuels that contribute significantly to environmental degradation. Integrating PV power plants into the energy portfolio helps diversify energy sources, thereby reducing dependency on oil and natural gas.

• Reducing GHG Emissions: Transitioning to renewable energy aligns with global efforts to combat climate change. PV power plants produce electricity without emitting GHGs, supporting Iraq's commitment to reducing its carbon footprint.

• Meeting Growing Energy Demand: Iraq faces rising electricity demand due to population growth and urbanization. PV power plants can supplement existing power generation, ensuring a stable and reliable energy supply.

• Economic Benefits: Developing PV projects stimulates local economies by creating jobs for installation, operation, and maintenance. It also reduces expenditure on fuel imports for electricity generation, freeing resources for other developmental needs.

• Energy Security: Solar energy reduces vulnerabilities associated with the fluctuating availability and prices of fossil fuels. By investing in PV infrastructure, Iraq can enhance its energy independence and resilience.

2.1 Utilization of Unproductive Land

It is well known that solar power plants require large areas of land. Iraq, with its vast regions of desert terrain, is a reality. Large tracts of unused desert land can be converted into productive solar farms, maximizing land use and contributing to sustainable development.

• Support from global and regional initiatives: International support and funding opportunities from organizations and countries promoting renewable energy provide a favorable environment for Iraq to advance its solar ambitions.

• Technological Advancements: Advances in PV technology have significantly reduced the cost of solar panels and improved their efficiency, making solar power an increasingly viable and attractive option.

• Improving energy Access in remote areas: PV power plants, particularly small-scale installations, can bring electricity to remote and underserved regions, improving living standards and supporting rural development.

By capitalizing on its solar potential, Iraq can achieve sustainable growth, enhance energy security, and contribute to global environmental goals while effectively addressing its energy challenges.

2.2 RETScreen Program for Solar Photovoltaic Systems

RETScreen is a clean energy management software developed by Natural Resources Canada. It is widely used for feasibility analysis, energy modeling, and performance monitoring of renewable energy systems, including PV power plants [20, 21]. Figure 1 illustrates the application of RETScreen for PV systems.

2.3 Key Features of the Proposed Solar Photovoltaic System

Feasibility Study: RETScreen enables users to evaluate the technical and financial feasibility of PV projects. It includes a comprehensive database of solar resource data, equipment specifications, and cost information. Users can assess factors like energy production, system sizing, and environmental impact, as shown in Figure 2.

Energy Production Estimation: The software uses location-specific solar irradiance data to estimate annual energy generation from a PV system. It accounts for panel orientation, tilt, shading, and temperature effects.

Financial Analysis: RETScreen limits parameters such as internal rate of return (IRR), net present value (NPV), and payback duration.

GHG Emission Reduction: RETScreen estimates the reduction in GHG emissions resulting from the transition to solar energy, supporting environmental goals.

Hybrid System Modeling: RETScreen supports the design and analysis of hybrid systems, combining PV with other renewable or conventional energy sources.

Performance Monitoring: RETScreen enables the tracking and analysis of actual performance data against expected results for operational PV systems, ensuring optimal operation [22, 23].

2.4 Steps to Analyze a Solar Photovoltaic Project in RETScreen

• Input Location and Climate Data: It can select the project location from the software's climate database or input specific solar irradiance data manually.

• Define System Specifications: Enter details about the PV system, including the type and size of panels, efficiency, degradation rates, and Inverter specifications.

• Set Financial Parameters: Include project costs (capital, operating, and maintenance), energy prices, and discount rates.

• Analyze energy performance: RETScreen calculates energy output based on system specifications and local climate conditions.

• Evaluate Financial and Environmental Impact: Review financial metrics and GHG emission reductions to determine project viability.

2.5 Advantages of Using RETScreen and Applications

The platform enables users to assess the viability, performance, and environmental impact of energy technologies with high accuracy and flexibility. One of the key advantages of RETScreen is its ability to perform pre-feasibility and feasibility analyses by integrating technical, financial, and environmental data in a user-friendly interface. It supports various technologies, including solar, wind, hydro, bioenergy, and geothermal. RETScreen includes advanced features such as performance analysis and benchmarking, emissions reduction tracking, and risk assessment. Due to its adaptability and free access, RETScreen is widely used by researchers, engineers, project developers, and policymakers worldwide. Its applications extend across educational, industrial, and governmental sectors, making it an essential tool for planning and managing sustainable energy projects. This subsection reports the benefits of using RETScreen and highlights case studies that demonstrate its practical utility in real-world scenarios.

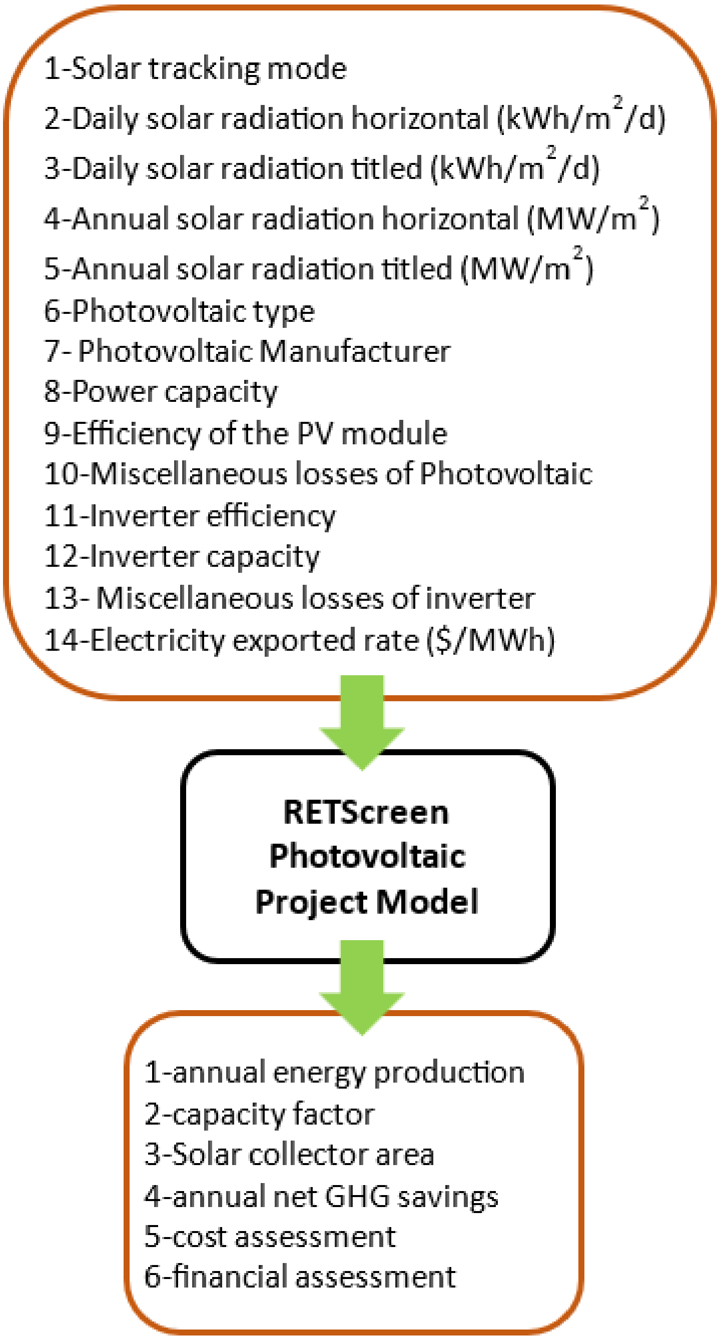

Figure 1. RETScreen software

- Simplifies complex analyses.
- Comprehensive Database: Provides access to global climate data and equipment specifications.
- Versatile Applications: Supports various project sizes, from residential to utility-scale systems.
- Cost-Effective: Reduces the need for third-party feasibility studies, saving time and resources.
- Supports Decision-Making: Offers insights for policymakers, investors, and engineers.

And applications,

- Pre-feasibility and feasibility studies for new solar installations.
- Retrofitting existing systems with upgrades or expansions.
- Designing hybrid systems combining PV with other energy sources.

• Tracking operational performance and optimizing energy production.

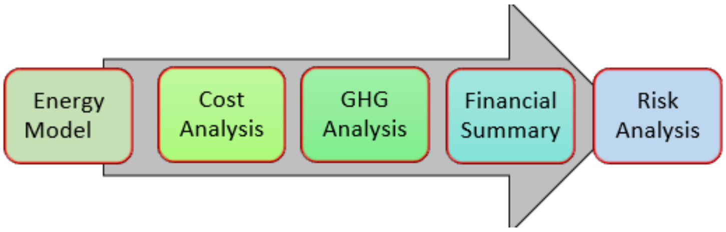

Figure 2. The RETScreen program steps through standard project analysis

2.6 Identifying Sites for Establishing Solar Photovoltaic Energy Plants and Site Data

Figure 3 shows the identification of two sites in the governorates of Al Sulaymaniyah and Al Muthana to conduct a technical and financial feasibility study for establishing solar PV power plant projects to produce electricity in Iraq.

Figure 3. Locations of the two proposed solar photovoltaic project sites in Iraq

Table 1 illustrates the climate data for the two identified sites. Figure 4 shows that the climate changes throughout the year, and the maximum daily solar radiation can be obtained from April to September.

Table 1. Climate data for both sites

Item	Al Sulaymaniyah Project	Al Muthana Project
Longitude	45.45	45.279
Latitude	35.549	31.319
Annual solar radiation–horizontal	4.9478 $\mathrm{kWh/m^2}$·day	5.1957 $\mathrm{kWh/m^2}$·day
Annual solar radiation–tilted	5.4084 $\mathrm{kWh/m^2}$·day	5.5207 $\mathrm{kWh/m^2}$·day

The SunPower (SPR-E20-435-COM) PV panels are suitable for a wide range of solar energy systems, including both grid-tied and off-grid configurations. They can also be integrated with energy storage units, such as batteries, allowing users to store surplus electricity for use during periods of peak demand or power failures. Table 2 shows the technical specifications of SunPower (SPR-E20-435-COM) solar panels used in this analysis.

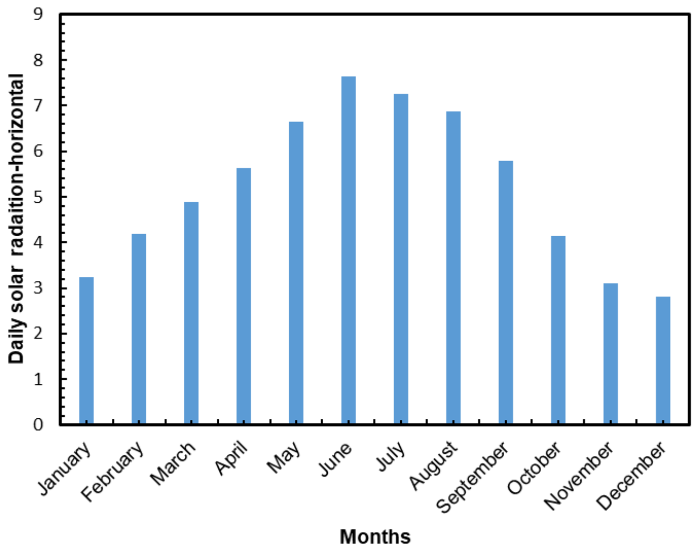

Figure 4. Monthly average solar radiation data

Table 2. Specification of SunPower (SPR-E20-435-COM) solar panels

Item	Specification
Brand, model	SunPower, SPR-E20-435-COM
Wattage, efficiency	435 Watt, 20.3%
Country of manufacturer	US
Type	Monocrystalline
Power tolerance	+5% or −3%
Short-circuit current (ISC), Open-circuit voltage (VOC)	6.43 A, 85.6 V
Temperature coefficient	0.42
Temperature	−40 °C to +85 °C
Output warranty	25 yr
Length, width, depth, weight	2,067 mm; 1,046 mm; 46 mm; 25.4 kg
Max. load	Wind: 50 psf, 2,400 Pa, 244 kg/m^2 front & back; Snow: 112 psf, 5,400 Pa, 550 kg/m^2 front
Junction box	IP-65; 1,230 mm cables/PV4S
Standard tests	IEC 61215, IEC 61730, UL1703 (Type 2 Fire Rating)
Frame	Class 2 silver anodized stacking pins
Tempered glass	High-transmission tempered anti-reflective
Appearance	Class A
Impact resistance	25 mm diameter hail at 23 m/s

3 Results and Discussion

3.1 Technical Results

This analysis outlines key details of the energy projects, including their locations, the processes types and technologies used in each project, energy loads, and the renewable energy sourcing. The RETScreen software then uses the provided information to determine annual energy production and possible savings. Additionally, the worksheet captures data on solar resources and energy loads. It is typically accompanied by one or more sub-worksheets, such as an equipment data sheet, to support the energy model. Table 3 shows the technical results of establishing solar PV power projects for both sites, especially the electricity exported to the grid.

3.2 Emission Analysis

The GHGs Emission Reduction Analysis Model' is build-in feature in the GHGs Analysis worksheet of the RETScreen software. This feature supports the software users to predict the emission reduction potential of a proposed clean energy project. This standardized model, used across all RETScreen "Clean Energy Technology

Models", calculates the GHG emission profiles for both the base case system and the proposed clean energy case system. The potential reduction in emissions is determined by comparing the GHG emission factors and incorporating other key data generated by RETScreen, such as annual energy output, as shown in Table 4 for both proposed projects.

Table 3. Technical results for both sites

Item	Solar Photovoltaic (PV) Power Plan Project	
	Al Sulaymaniyah	Al Muthana
Field general data		
Solar tracking mode	Fixed	Fixed
Azimuth	0	0
Slope	36	31
PV panel		
Brand	Sunpower-mono-Si-SPR-E20-435-COM	Sunpower-mono-Si-SPR-E20-435-COM
Number of units	92,000	92,000
Efficiency	20.7%	20.7%
Solar collector area	193,333	193,333
Miscellaneous losses	15%	15%
Inverter		
Inverter efficiency	98%	98%
Inverter capacity	32,000 kW	32,000 kW
Inverter Miscellaneous losses	1%	1%
Capacity factor	17.5%	17.2%
Electricity is exported to the grid (kWh)	61,238,135	60,321,186

Table 4. Estimated greenhouse gas (GHG) emission reduction for both sites

Item	Al Sulaymaniyah Project	Al Muthana Project
Base case [using fossil fuel]	59,482.2 tCO_2	58,591.5 tCO_2
Proposed case [solar photovoltaic]	4,163.7 tCO_2	4,101.4 tCO_2
Gross annual reduction in GHG emission	55,318.4 tCO_2	54,490 tCO_2
Gross annual reduction in GHG by non-use of cars and light trucks	10,131.6 tCO_2	9,979.87 tCO_2
Gross annual reduction of crude oil consumption	128,647.5 barrels	126,721.2 barrels
Gross annual forest area saved for absorbing CO_2	12,572.4 acres	12,384.1 acres

Some encounters may occur in the calculations associated with a GHG analysis. The following general parameters have been input to the RETScreen software in addition to specific parameters of the base case/proposed cases: country (Iraq), fuel type (all types), and T and D losses (7%). Figure 5 shows the GHG emission reduction analysis and compares the project's base case and proposed cases, where the base case (likely conventional fossil-based systems) and the proposed case (a PV technology) are considered.

For the proposed cases, the user records the initial, annual, and periodic costs, along with any credits from base case costs that are excluded. Alternatively, incremental costs could be entered into directly. The user can choose between conducting a pre-feasibility study or a full feasibility study. An elementary feasibility analysis requires less detailed and less precise data, while a feasibility analysis demands more accurate and comprehensive data. This is because the calculations performed by the RETScreen software at this stage are relatively simple, involving basic addition and multiplication.

• The initial cost of establishing each proposed project: $39,619,800.

• O&M cost savings for the year for each proposed project: $440,220.

Model of Financial Analysis: The RETScreen Financial Analysis Model, located in the Financial Summary worksheet of the software, enables users to enter key financial parameters such as discount rates and inflation. It then automatically calculates important financial indicators, including IRR, simple payback (SPB) period, and NPV. This section also outlines the equations used within the model. The inflation rate (%) represents the estimated average annual inflation over the project's lifetime. For instance, inflation in Iraq over the next 25 years is projected to range between 2% and 3%.

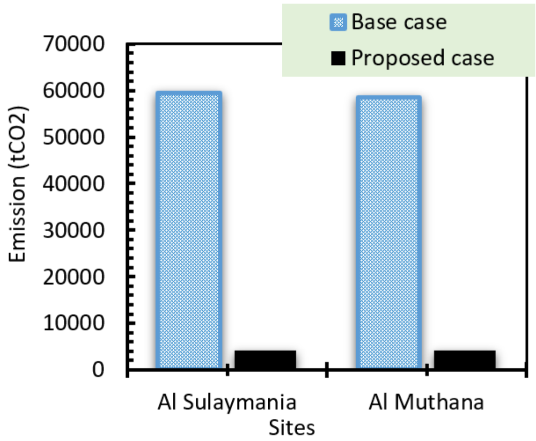

Figure 5. Emission analysis

The discount rate (%) is the rate applied to future cash flows to determine their present value. It is often referred to as the "hurdle rate," "cut-off rate," or "required rate of return" when evaluating a project's financial viability. Typical discount rates range from 7% to 11%. Monetary incentives and grants refer to contributions, subsidies, or grants provided to offset the initial project cost (excluding credits). In the RETScreen model, these incentives are considered non-refundable and are treated as revenue during the development or construction phase (year 0).

• Fuel cost increment: 2%;
• Rate of Inflation: 3%;
• Rate of Reinvestment: 9%;
• Project lifetime: 25 years;
• Debt ratio: 70%;
• Debt interest ratio: 6%;
• Debt term: 15 years.

The financial analysis of the (IRR, SPB, NPV, annual life cycle savings) account is based on several scenarios and according to different definitions, as follows:

• Scenario-1 (Sc1): Implementation of the projects in case grants = \$0 & GHG reduction credit rate = \$0/tCO_2 (tCO_2 = tonne of carbon dioxide).

• Scenario-2 (Sc2): Implementation of the projects in case grants are 50% of the capital and the GHG reduction credit rate is \$0/tCO_2.

• Scenario-3 (Sc3): Implementation of the projects in case grants = \$0 & GHG reduction credit rate = \$25/tCO_2 with GHG reduction credit duration = 10yr (If a 10-year crediting period is selected, Certified Emission Reductions can be certified and issued for the entire duration of the project once it has been validated and registered, without the need to reassess the baseline during that period).

If applicable, the GHG reduction credit is calculated per tCO_2 and is used in conjunction with the net GHG reduction to determine the annual revenue from GHG reduction.

3.3 Internal Rate of Return

The IRR on equity (%) represents the accurate interest yielded on the project's equity over its lifetime before income tax. It is based on the pretax annual cash flows and the project duration. Also known as the Return on Equity, Return on Investment, or Time-Adjusted Rate of Return, it is determined by identifying the discount rate that makes the NPV of the equity equal to zero. Therefore, knowing an organization's discount rate is not required to use this indicator. Instead, organizations can compare the IRR to their required return, often equivalent to their cost of capital, to assess the project's financial attractiveness. The IRR is calculated on a nominal basis (i.e., including inflation), as shown in Table 5 and Figure 6.

The SPB period (in years) indicates how long it takes for a proposed project to recover its initial investment through the revenue or savings it generates. The core idea behind the SPB method is that investments with shorter payback periods are generally more attractive. For instance, in an energy project, a negative payback period indicates that the annual operating costs exceed the yearly savings, rendering the project financially unviable. Table 6 shows the SPB period for establishing both proposed projects versus the tariff, as shown in the graph in Figure 7.

Table 5. Estimated internal rate of return (IRR) versus tariff for both projects

Tariff ($/kWh)	IRR (%)					
	Al Sulaymaniyah Project			Al Muthana Project		
	Sc1	Sc2	Sc3	Sc1	Sc2	Sc3
0.015	−13.6	−12.4	−11.6	−13.9	−12.8	−11.9
0.035	0.4	Positive	4.7	0.2	Positive	4.3
0.055	9.2	Positive	15.6	8.8	Positive	15
0.075	17.7	Positive	26.1	17.2	Positive	25.3
0.095	26.9	Positive	36.6	26.2	Positive	35.7

Note: Sc1 = Scenario-1, Sc2 = Scenario-2, Sc3 = Scenario-3.

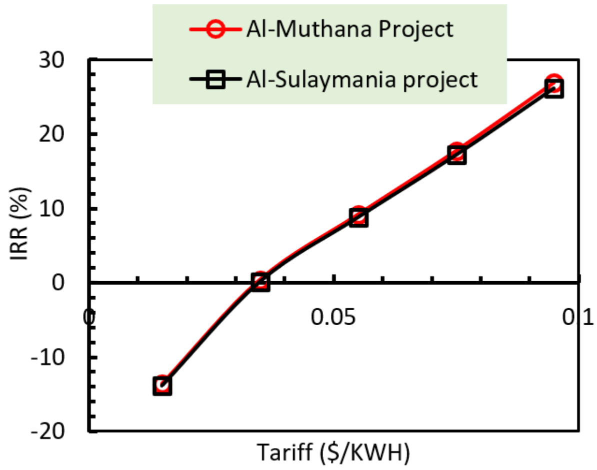

Figure 6. Internal rate of return (IRR) on equity (%) versus tariff

Table 6. Simple payback (SPB) versus tariff for both projects

Tariff ($/kWh)	SPB (years)					
	Al Sulaymaniyah Project			Al Muthana Project		
	Sc1	Sc2	Sc3	Sc1	Sc2	Sc3
0.015	82.8	41.4	21.3	85.3	42.6	21.7
0.035	23.3	11.6	12.8	23.7	11.8	13.1
0.055	13.5	6.7	9.2	13.8	6.9	9.3
0.075	9.5	4.8	7.2	9.7	4.8	7.3
0.095	7.4	3.7	5.9	7.5	3.7	5.9

Note: Sc1 = Scenario-1, Sc2 = Scenario-2, Sc3 = Scenario-3.

Table 7 and Figure 8 present the NPV of the projects, which represents the total value of future cash flows discounted to today's currency using a specified discount rate. Closely related to the IRR, the NPV is calculated at time zero, marking the transition from the end of year 0 to the start of year 1. Under the NPV method, the present value of all projected cash inflows is compared to the present value of all projected outflows for a given investment. The resulting difference, known as the NPV, indicates the project's financial viability: a positive NPV suggests the project is potentially profitable and acceptable from an investment standpoint. To apply this method, a discount rate must be selected to convert future cash flows into present value. Organizations typically invest significant effort into determining an appropriate discount rate. The RETScreen model calculates NPV using cumulative after-tax cash flows. However, if the user opts out of tax analysis, the NPV will reflect pretax cash flows instead.

Table 8 and Figure 9 display the annual life cycle savings ($/year), which represent the levelized nominal annual savings that have the same duration and NPV as the project. These savings are calculated based on the project's NPV, discount rate, and lifespan, providing a consistent annual figure that reflects the overall financial benefit over the project's life.

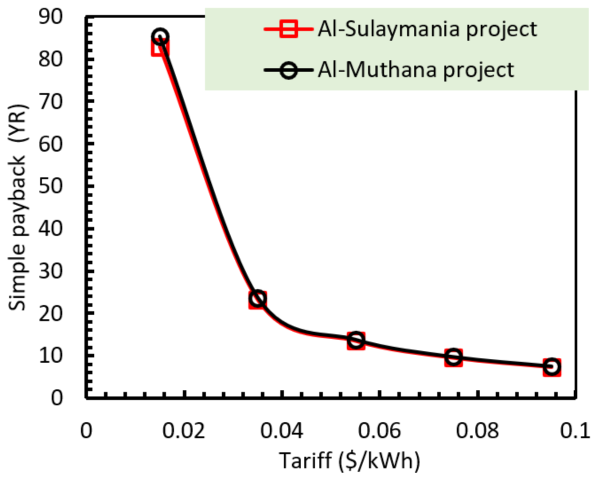

Figure 7. The simple payback (year) versus tariff

Table 7. Net present value (NPV) versus tariff for both projects

Tariff ($/kWh)	NPV ($)					
	Al Sulaymaniyah Project			Al Muthana Project		
	Sc1	Sc2	Sc3	Sc1	Sc2	Sc3
0.015	−29,472,201	−9,662,301	−20,596,834	−29,639,210	−9,829,310	−20,896,739
0.035	−14,600,682	5,209,217	−5,725,316	−14,990,370	4,819,530	−6,247,899
0.055	270,836	20,080,736	9,146,202	−341,531	19,468,369	8,400,941
0.075	15,142,354	34,952,254	24,017,720	14,307,309	34,117,209	23,049,780
0.095	30,013,872	49,823,772	38,889,238	28,956,149	48,766,049	37,698,620

Note: Sc1 = Scenario-1, Sc2 = Scenario-2, Sc3 = Scenario-3.

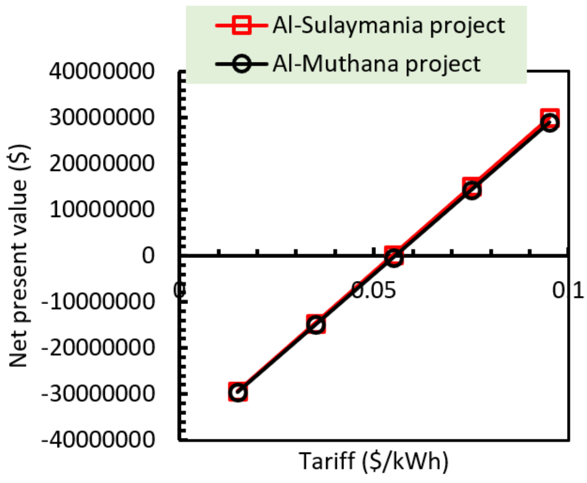

Figure 8. Net present value ($) versus tariff

3.4 Risk Analyses

Risk analysis depends on the uncertainty associated with several key input parameters and evaluating the influence of this uncertainty on pretax IRR-equity, pretax IRR-assets, after-tax IRR-equity, after-tax IRR-assets, equity payback, NPV or Energy Production Cost: In the risk analysis section, the influence of each input parameter on a financial indicator is assessed using a standardized multiple linear regression model. This approach quantifies how variations in key inputs affect economic outcomes, such as NPV or the cost of energy production, as shown in Table 9.

Table 8. The Annual Life-cycle Savings Under Different Electricity Tariffs

Tariff ($/kWh)	The annual life cycle savings ($/yr)					
	Al Sulaymaniyah Project			Al Muthana Project		
	Sc1	Sc2	Sc3	Sc1	Sc2	Sc3
0.015	−3,000,455	983,683	−2,096,886	−3,017,457	1,000,685	2,127,419
0.035	−1,486,441	530,331	−582,873	−1,526,113	490,658	−636,075
0.055	27,573	2,044,344	931,141	−34,770	1,982,002	855,268
0.075	1,541,587	3,558,358	2,445,154	1,456,573	3,473,345	2,346,612
0.095	3,055,600	5,072,371	3,959,168	2,947,917	4,964,689	3,837,955

Note: Sc1 = Scenario-1, Sc2 = Scenario-2, Sc3 = Scenario-3.

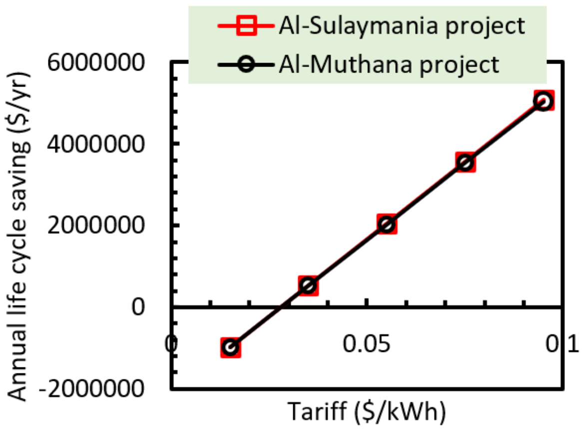

Figure 9. The annual life cycle savings ($/yr) versus the tariff

Table 9. Risk analysis for both projects

Parameter	Unit	Value	Range	Minimum	Maximum
Initial costs	$	39,619,800	25%	29,714,850	49,524,750
O&M	$	440,220	25%	330,165	550,275
Electricity is supplied to the Sulaymaniyah site grid	MWh	61,238.1	25%	45,928.6	76,547.7
Electricity is supplied to Al Muthana site grid	MWh	60,321.2	25%	45,240.9	75,401.5

4 Conclusions

The establishment of solar PV power plant projects in Iraq represents a significant step toward a more sustainable and secure energy future for the country and the scalability of solar energy initiatives in the region. Where it is possible, through the establishment of solar PV power plants and with the help of grants and government support for these projects, to fill the electricity shortage resulting from the population growth in Iraq, where the regions can be supplied with a capacity more on 61,238,135kWh and by imposing an appropriate tariff higher than $0.035/kWh to achieve the appropriate feasibility of the project with a SPB less than 11.6 yr and an appropriate rate NPV more than $5,209,217 and this encourages the reduction of GHGs annually to 55,318.4 tCO_2 for Al Sulaymaniyah project and 54,490 tCO_2 for Al Muthana project.

Author Contributions

Conceptualization, H.A.A.W. and L.F.A.M.; methodology, H.A.A.W.; software, L.F.A.M.; validation, M.E.A. and F.K.K.; formal analysis, F.K.K.; investigation, L.F.A.M.; resources, L.F.A.M. and F.K.K.; data curation, H.A.A.W.; writing—original draft preparation, L.F.A.M.; writing—review and editing, H.A.A.W.; visualization, F.K.K.; supervision, M.E.A. All authors reviewed and approved the final version of the manuscript.

Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

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Conflicts of Interest

The authors declare no conflicts of interest.

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